PROBLEM STATEMENT CODE: PS000109 • DIGITAL OCCUPATIONAL HEALTH MONITORING

WorkCheck Kachchh (કચ્છ શ્રમિક સુરક્ષા)

CORE MISSION: Find exposed informal workers via **public health clinics**, trigger check-ups via **₹2,015 solar sensors**, and track referrals into an **ABHA Worker Health Pass.**

STEP 1 • SCREEN
1-Min Clinic Check
Past Jobs & Symptoms

→

STEP 2 • SENSE
₹2,015 Solar Node
Heat (WBGT) & Dust Alert

→

STEP 3 • EXAMINE
BOCW Mobile Van
Free Doctor Check-Up

→

STEP 4 • OWN
e-Shram + ABHA Pass
Portable QR Health Record

TEAM ID	TRACK CATEGORY	INSTITUTION / ORGANIZATION	TEAM LEADER
TM005024	College Track (Engineering)	LDRP Institute of Technology & Research, Gandhinagar, Gujarat	Jay Patel (Team Leader)

CHALLENGE DEFINITION

Problem Statement & Real-World Context

Focus: Root Cause & Citizen Friction

Root-Cause Flowchart & Field Evidence

1.8 LAKH

1. Unregistered Informal Laborers

Salt pans, quarries, ports & cement off factory rolls

▼

AGE 45+/40+

2. Outdated Age-Only Check-Up Rules

Rule 9 & 5 miss younger workers harmed by daily dust/heat

▼

87.5% HEAT

3. Extreme Heat & Silica Dust Harm

2.1× kidney risk • 26% > silica limit • 2,067 hot hrs (Kandla)

▼

15% AUDIT

4. Unrecorded Disease & Wage Fear

182 silicosis cases & 80 deaths (PTRC) vs 562 logged nationally

4 Target Groups & Why Market Fails

① Agariya Salt Rakers
Extreme Sun & Brine Glare

② Stone Quarry Drillers
Respirable Silica Lung Dust

③ Port & Dock Loaders
Kandla & Mundra Marine Heat

④ Mobile Van Doctors
BOCW Dhanvantari Rath Queue

WHY EXISTING MARKET SOLUTIONS FAIL INFORMAL LABOR:

₹4,100 / Mo

US Wearable Patches (Kenzen, SlateSafety)
560× too costly for daily-wage workers

₹5.5 Lakhs

Lab Dust Monitors (TSI DustTrak)
Measures air only — zero link to doctor referrals

Payroll Only

Corporate EHS Software (OccuCare)
Tracks factory staff — misses migrating labor

SOLUTION APPROACH •

Proposed Solution & Innovation Novelty

Ethos: Frugal, Accessible, Scalable

END-TO-END WORKFLOW DIAGRAM • WORKS 100% EVEN WITHOUT HARDWARE (SENSOR-OPTIONAL)

1. PRIORITIZE	2. 1-MIN CHECK	3. 3-RULE TRIAGE	4. VAN EXAM	5. KEEP RECORD
Heat & Dust Map Rank High-Risk Sites	**Clinic / Volunteer** Log Job Years & Symptoms	**Auto Risk Engine** Age • Past Dust • Symptoms	**BOCW Mobile Van** Doctor X-Ray & Creatinine	**QR + ABHA Locker** Portable Across Employers

Core Novelty

Portable QR Health Pass
12-Digit e-Shram UAN + ABHA ID follows worker

Past Exposure Memory
Counts career dust/heat years (e.g., 20 yrs at crusher)

100% DPDP Act Privacy
Worker OTP lock • Anonymous DDO taluka heatmaps

Technical Moat

150mm Black Copper Globe
True ISO 7243 solar radiant heat (zero wind error)

Coastal Humidity Math Fix
Prevents sea-air false dust spikes ($R^2 > 0.88$)

1-Click ABDM FHIR Sync
Instant HL7 FHIR R4 report deposit to ABHA

"Seva First" Value

< ₹7.30 / Worker / Month
₹0 worker cost — 560× cheaper than US wearables

100% Cess & DMF Funded
Backed by Gujarat BOCW 1% Cess & PMKKKY

Gujarati • Hindi • English
Native trilingual UI (ગુજરાતી) + RGB flag alerts

LIVE SANDBOX & DEMO

Interactive 4-Persona Working Prototype (DDO, Van Doctor, Supervisor & Worker Views):
https://jaypatel29042008-glitch.github.io/WorkCheck-Kachchh-SSIP/

SYSTEM ARCHITECTURE •

Technical Architecture || Methodology

End-to-End System Pipeline

① EDGE SENSING LAYER
Worksite Hardware (Teal)

SOLAR IOT NODE	**₹2,015 Bulk BOM • ESP32** Installed at salt pans, quarries & ports
150mm BLACK GLOBE	**True Radiant Heat (Tg)** Eliminates wind error ($R^2 = 0.92$)
DUST + CLIMATE + UV	**PMS5003 • BME280 • UV** Laser PM2.5/PM10, Temp/RH & Glare
RGB FLAG LAMP	**4-Color Work-Rest Light** Green / Yellow / Orange / Red on site
ZERO-SENSOR MODE	**100% Software Fallback** Clinic screening works without hardware

② COMPUTATION & RULES
FastAPI + SQL Engine (Orange)

ISO 7243 WBGT MATH	**0.7 Tnw + 0.2 Tg + 0.1 Ta** ACGIH heavy-work heat strain index
COASTAL DUST FIX	**$PM_{raw} / [1+0.25 \cdot RH^2/(1-RH)]$** Prevents sea humidity fogging
10-TABLE SQL DB	**30-Day Raw + Rollups** Longitudinal worker exposure history
3 ROUTING RULES	**1. Age 40/45+ • 2. Quartz Yrs** 3. Symptoms (Dark urine, cough)
DOCTOR OWNERSHIP	**Clinician-Reviewed Rules** Software flags risk; van doctors diagnose

③ DELIVERY & GOVERNANCE
Worker, Van & DDO (Emerald)

WORKER QR PASS	**12-Digit e-Shram UAN** Portable bilingual card across contractors
BOCW MOBILE VAN	**Dhanvantari Arogya Rath** Spirometry, Creatinine & ILO X-Ray
1-CLICK ABHA SYNC	**HL7 FHIR R4 Publisher** Deposits reports into worker Health Locker
AUTO FORM 5 CERT	**Gujarat OSH Rule 9** Annual statutory fitness certificate
DDO GIS HEATMAP	**Anonymous Taluka Map** 1-Click BOCW / NPCCHH / ESIC CSVs

ENGINEERING STACK •

Technology Stack & Feasibility

Component Readiness & Dependencies

A. EMBEDDED IOT & HARDWARE

TRL-5 • VALIDATED

ESP32 Dual-Core
240MHz + Offline Flash Buffer

BME280 + PMS5003
Laser Dust, Temp/RH & UV

150mm Black Globe
Copper Float + 10k Thermistor

₹2,015 Unit BOM
5V Solar + Li-ion + IP65 Box

B. BACKEND & RULES ENGINE

TRL-6 • 21/21 PASS

Python 3.10 + FastAPI
Async REST + OpenAPI Docs

10-Table ANSI SQL
30-Day Raw + Career Rollups

ISO 7243 + Aerosol
WBGT & Humidity Correction

21/21 Tests Passing
Automated Integration Suite

C. FRONTEND & CLINICAL UI

TRL-6 • LIVE PWA

Offline-First PWA
Mobile-First Tailwind CSS

Leaflet GIS + Chart.js
Live Taluka Heatmaps

Gujarati • Hindi • EN
Native Trilingual UI (ગુજરાતી)

4 User Personas
Worker, Site, Van Doctor, DDO

D. STANDARDS & PRIVACY

TRL-6 • COMPLIANT

ABDM HL7 FHIR R4
1-Click DiagnosticReport Sync

12-Digit e-Shram UAN
National QR Health Pass

10 Public Datasets
ERA5, CAMS, CPCB, MSHA, OSHA

100% DPDP Act 2023
Worker OTP + Anonymous GIS

IMPACT ASSESSMENT • SIH SECTION 05

Socio-Economic Impact & Viability

Quantifiable Citizen Outcomes

1. CLINICAL IMPACT

>90% Follow-Up

2.1× Kidney Risk Addressed

KIDNEY PROTECTION

Prevents AKI & CKDu

87.5% salt-pan heat stress & 2.1× kidney risk

SILICOSIS CAPTURE

Early Lung Screening

Flags quartz dust years before lung fibrosis

ZERO LEAKAGE

>90% Referral Tracking

Follows every case to clinic vs 15% audit

2. FRUGAL ECONOMICS

₹7.30 / Month

560× Cheaper Than Wearables

560× CHEAPER

₹7.30 vs ₹4,100 / Mo

Replaces costly US patches (Kenzen)

₹2,015 SITE NODE

Replaces ₹5.5L DustTrak

1 solar node covers entire worksite

₹2.97L PILOT

50 Sites • 1,000 Workers

Year-1 district budget (₹88/worker/yr)

3. PRODUCTIVITY

~358 Hrs/Yr

~20% Severe Heat Avoided

6 AM – 3 PM SHIFT

Cool-Shift Optimization

Avoids ~20% peak heat (~358 hrs/yr)

RGB REST LAMPS

Zero Employer Disputes

Objective ISO 7243 work-rest light

AMENITY AUDIT

Water & Shade Tracking

Verifies drinking water & rest tents

4. GOVERNANCE

₹0 Worker Cost

100% BOCW Cess & DMF Funded

100% FREE

₹0 Paid by Laborers

Health data is never sold or monetized

CESS FUNDABLE

BOCW 1% Cess & DMF

Uses existing state welfare budgets

RULE 9 & 5 READY

Auto Form 5 & Heatmaps

Instant compliance for DDO Kachchh

EXECUTION TIMELINE • SIH SECTION 06

Implementation Roadmap & Milestones

From Prototype to Field Rollout

STAGE 1: LAB VALIDATION

Oct – Nov 2026 • TRL-5
STATUS: COMPLETED

1. Datasets Analyzed
ERA5, CAMS, CPCB & OSHA (5 Sites)

↓

2. Core Engine Built
10-Table SQL + ISO 7243 Rules

↓

3. 4-Persona Web App
1-Click ABHA FHIR Publishing

↓

4. Verified & Live
21/21 Tests • GitHub Pages Live

STAGE 2: PILOT FIELD TRIAL

Nov – Dec 2026 • TRL-6
STATUS: PLANNED

1. Field Site Nodes
2–3 Solar Nodes (Rapar & Abdasa)

↓

2. Lab Calibration
Validate vs Reference Instruments

↓

3. PHC / CHC Pilot
1-Min Exposure Check at 1 Clinic

↓

4. Clinical Sign-Off
CDHO Kachchh & NIOH Review

STAGE 3: VAN INTEGRATION

Jan – Mar 2027 • TRL-7
STATUS: PLANNED

1. Mobile Van Fleet
BOCW Dhanvantari Rath Link

↓

2. 50-Site Rollout
Cover 1,000 Informal Workers

↓

3. QR Health Passes
Bilingual e-Shram PVC Cards

↓

4. Live ABHA Sync
Spirometry & Creatinine Deposits

STAGE 4: DISTRICT SCALE

Apr 2027+ • TRL-8
STATUS: TARGET

1. All 10 Talukas
Scale to 10,000+ Workers

↓

2. ESIC Dispensaries
Bhuj & Gandhidham Referrals

↓

3. SEWA Insurance
Parametric Heat Payout Trigger

↓

4. District Policy
Anonymized DDO GIS Reports

SQUAD CAPABILITIES

Multi-Disciplinary Team & Mentorship

Complementary Skill Sets

TL

Jay Patel

Team Leader & Architect

- **3-Layer Architecture**
  End-to-End Platform Design
- **10-Table SQL & Rules**
  ISO 7243 & Aerosol Engine
- **ABDM HL7 FHIR R4**
  1-Click ABHA Locker Sync
- **21/21 Test Harness**
  Automated CI Verification

M2

Hardware & IoT Lead

Embedded & Calibration

- **ESP32 Dual-Core**
  Firmware & Wokwi Model
- **150mm Black Globe**
  Radiant Heat Calibration
- **Sensor Array Wiring**
  PMS5003, BME280 & UV
- **₹2,015 Solar BOM**
  5V Solar + Li-ion Design

M3

Public Health Lead

Field Ops & Localization

- **12-Digit e-Shram QR**
  Portable Worker Health Pass
- **BOCW Mobile Van**
  Dhanvantari Rath Liaison
- **Gujarati • Hindi • EN**
  Trilingual UI (ગુજરાતી)
- **1-Min Clinic Triage**
  ASHA & ANM Workflow

M4

Compliance Lead

Hygiene & Data Privacy

- **Gujarat OSH Rule 9**
  Auto Form 5 Certification
- **Factories Act 41C**
  Central Rule 5 Mapping
- **DPDP Act 2023**
  OTP Lock & Anonymous GIS
- **10 Public Datasets**
  Empirical Hazard Modeling

Faculty Mentor: Prof. / Dr. Faculty Mentor

Department of Computer Engineering / IT, LDRP-ITR, Gandhinagar

Ready for Jury Review & Evaluation